Hard-Soft Acid-Base Theory

Lishan Prado

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1 Introduction

Hard-Soft Acid-Base Theory, or HSAB, is a very interesting qualitative way to judge the tendencies for two things to react. I'll begin by introducing the fundamentals and how it works, and then I'll tell you about some of the many cool applications where HSAB pops up.

2 How It Works

HSAB hinges on the idea of Lewis acids and Lewis bases. You may remember the three possible definitions of acids/bases, and the Lewis definition is the one centering around the movement of electrons.

Lewis acids accept electron pairs, while Lewis bases donate electron pairs. Based on these definitions, it should make sense that Lewis acids tend to be cations, while Lewis bases are anions.

Let's now look at what "hard" and "soft" mean. The idea of "hardness" or "softness" has nothing to do with actual texture or anything physical. In layman's terms, it is just a measure of "charge density". If a species is hard, it means that it has high charge density and low polarizability, while if it's soft, it has low charge density and high polarizability. As a consequence of these rules, small size, low polarizability, and/or high charge tends to imply hardness, while large size, high polarizability, and/or low charge tends to imply softness.

Let's look at Al^{3+}. This has a lot of charge (magnitude of 3), and it's a relatively small ion, so this is a classic hard species. Also note that it has a positive charge, meaning it would much rather accept electrons than to donate them, so this is a hard acid.

Now let's look at iodide I^-. This doesn't have a lot of charge (magnitude of 1), and it's a pretty large anion, so this is a soft base.

Note that classifying species as hard or soft is qualitative, so it's hard to definitively say if something is hard or soft. However, it is often possible to compare two species and determine which is harder and which is softer. For some practice, try looking at the following pairs of species and determining which is harder and which is softer.

2.1 Practice Problems

For each pair, determine which species is harder and which species is softer.

1. F^- and I^-

2. Ag^+ and Al^{3+}

3. OH^- and S^{2-}

4. Fe^{3+} and Fe^{2+}

5. Hg^{2+} and Mg^{2+}

2.2 Answers

Try to answer the five questions above before you check the answers below.

1. F^- is harder because while the bases have the same charge, fluoride is much smaller, so it has a higher charge density.

2. Al^{3+} is harder because it has a higher charge than the silver ion (+3 vs +1), and it is also smaller.

3. OH^- is harder than sulfide because the negative charge in the donor atom of hydroxide is on oxygen, which is much smaller than sulfur. This difference in size between oxygen and sulfur, as well as sulfur's high polarizability, offsets the larger charge (-2) on the sulfur atom compared to the charge on the oxygen atom (-1).

4. Fe^{3+} is harder because it has a higher charge (+3 vs +2).

5. Mg^{2+} is harder because it is smaller, and they have the same charge.

Now, what's the point? What's the point of classifying these ions into hard acids, soft bases, etc? Well, the whole point of HSAB is the idea that: hard acids bind better with hard bases, and soft acids bind better with soft bases.

Let's compare AgF and AgI. Silver is a pretty soft acid, overall, due to its low charge (+1) and relatively large size. So the question is: Which of two bases F^- and I^- is softer? Well, as you know by now, iodide is softer. Therefore, by HSAB, AgI is a more stable compound than AgF. As it turns, we have data to support this. AgI is a (yellow) precipitate, while AgF is famously soluble in water, proving that AgI is much more stable than AgF.

This is the entire point of HSAB. By this point, you know more than 90% of what you need to know about HSAB. I'll now talk about some interesting applications.

3 Applications

3.1 Minerals

If you've studied Earth science, you'll be familiar with several minerals. It turns out that we can predict whether certain metals are found in nature as oxides are as sulfides.

As you know, oxide O^{2-} is harder than sulfide S^{2-}. Therefore, harder metals are more likely to be found as an oxide. Corundum, for example, has formula Al_2O_3, which makes sense given Al^{3+} is a hard acid and bonds to the hard base O^{2-}. The corresponding sulfide, Al_2S_3, does not exist naturally, although it can be synthesized in a lab. This matches our understanding through HSAB perfectly.

Figure 1: Corundum

Another mineral called cinnabar consists of the compound mercury (II) sulfide, or HgS. This also makes sense, since mercury's large size causes it to be soft and preferentially bond to the soft sulfide anion. Interestingly, the mineral with formula HgO exists, although it is very rare. I had to google this, and I found that there is a rare mineral called montroydite, which is additionally extremely dense. The images of corundum and cinnabar were both found on their respective Wikipedia pages. I added them because I thought they looked cool.

Figure 2: Cinnabar

3.2 Linkage Isomerism

In metal complexes, the metal is bonded to several ligands. These ligands will almost always be Lewis bases, while the metal acts as a Lewis acid. One example is the stable $[Cu(H_2O)_6]^{2+}$, consisting of the relatively hard transition metal Cu^{2+} and the hard ligand H_2O, or water, which is hard due to oxygen's small size.

Some ligands are called ambidentate, which means they can bond through more than one atom. One example is the thiocyanate anion SCN^-, which can bond through either its sulfur atom or its nitrogen atom:

$$M{-}SCN$$

or

$$M{-}NCS.$$

HSAB helps us predict which of the two options a metal would prefer.

We'd have to compare the two atoms that the metal could bind to. The nitrogen atom is a harder base than the sulfur atom, so a harder metal would bond to the nitrogen, while a softer metal would choose the sulfur.

There are several other ligands that can choose which atom they want to use for bonding. For example, nitrite ${NO_2}^-$ can choose between the N and an O and cyanide CN^- can choose between the C and the N. Cyanate, OCN^-, also follows this pattern, and its opposite is called isocyanate, NCO^-.

$$M{-}OCN$$

or

$$M{-}NCO$$

The HSAB prediction says that the oxygen is harder, so a harder metal will choose cyanate and a softer metal will choose isocyanate.

There is also fulminate, which technically isn't an example of linkage isomer of cyanate and isocyanate, but is still worth discussing. Fulminate's formula is CNO^-, with the metal bonding to the carbon atom. The carbon is the softest of the three.

One example of a fulminate is $Hg(CNO)_2$. Of course, due to its large size, mercury (II) is soft, so it bonds reasonably well with the fulminate ion.

Fulminates are also generally reactive and explosive, since both cyanate and isocyanate are more stable than fulminate. In fact, the English language has a verb called "fulminate", which means to express loud or angry criticism, or to explode violently.

3.3 Solubility Rules

If you don't know the solubility rules of some random niche anion, or maybe you just forget your basic solubility rules, you can try using HSAB to intuit whether a salt is soluble.

I wrote a handout about Qualitative Cation Analysis which at one point mentioned that all 5th-period or 6th-period sulfides are insoluble. It's no coincidence that all the sulfides of this region are very insoluble, and it can be explained with HSAB. Sulfide is generally a soft base, so it bonds well with the large and soft acids of the 5th and 6th rows.

Another thing I mentioned in that handout is that fluorides often follow the opposite solubility rules of the rest of the halides. For example, silver chloride, silver bromide, and silver iodide are all insoluble. However, silver fluoride is soluble. As it turns out, fluoride is so small and electronegative that it is a very hard base, making its bonding tendencies very different than the rest of the halides.

3.4 α, β-Unsaturated Carbonyls

This section presumes a relatively strong sense of organic chemistry. If you are not at that point yet, feel free to skip this.

In organic chemistry, we call Lewis bases nucleophiles and Lewis acids electrophiles. Some hard nucleophiles include F^-, OH^-, OR^-, and R$-$MgBr. Some soft nucleophiles include I^-, CN^-, and R_2-CuLi. That last hard nucleophile is called a Gilman reagent, characterized by the copper atom. That last soft nucleophile is called a Grignard reagent, characterized by the magnesium atom. Gilman reagents tend to be softer than Grignard reagents because the C-Cu is more covalent in nature than the C-Mg bond. The C-Mg's ionic bond means carbon takes a larger negative partial charge than the carbon in the Gilman reagent.

Now let's look at an α, β-unsaturated carbonyl, also known as a Michael acceptor:

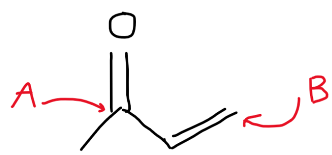

Figure 3: α, β-unsaturated carbonyl

This molecule has two electrophilic sites: the carbonyl carbon (A) and the carbon at the beta position (B). We can compare these two based on their hardness. The carbonyl carbon is closer to the electronegative oxygen, which will pull electron density from the carbon, making it more positively charged. This effect is much less prevalent for the carbon at position B. Therefore, the charge is larger for carbon A, so carbon A is the harder electrophile.

As it turns out, if you react a Grignard reagent with the Michael acceptor shown above, it will preferentially attack carbon A because the Grignard reagent is a hard nucleophile, and carbon A is a hard electrophile. At the same time, if you react a Gilman reagent with the same Michael acceptor, it will preferentially attack carbon B because the Gilman reagent is a soft nucleophile, and carbon B is a soft electrophile. For more information on this special organic chemistry idea, check Clayden's Organic Chemistry, 2nd edition, Chapter 22.

4 Limitations of HSAB

Although this handout has spent a while glazing HSAB, it is important to remember that it is not a perfect theory. It can generally predict trends of how certain acids and bases react, but it doesn't work for every compound.

One example is the solubility of alkaline earth metal fluorides. Fluoride is a hard base, so you'd expect BeF_2 to be very insoluble, with MgF_2 to be a little more soluble, followed by CaF_2, SrF_2, and BaF_2, maybe with BaF_2 even being soluble.

As it turns out, BeF_2 is very soluble in water, even though both species are very hard. As it turns out, solubility depends on more than just HSAB. For an ionic compound to dissolve, its lattice must break, and the resulting ions need to be stabilized by the water solvent molecules. The small size and high charge of Be^{2+} makes it strongly stabilized by water molecules, making the dissolution of BeF_2 much more favorable than you'd expect.

5 Conclusion

Overall, HSAB can be a very powerful tool, as we've seen from all the crazy things we can apply it to. I hope you learned something from this handout! I tried to make this as educational as possible, no matter your skill level. If you have questions or comments, or you see something wrong in this handout, or you want to tell me how much you love or hate this handout, feel free to contact me. Also, check out my website for more handouts that I've written.